

Racial Segregation and Black-White Longevity Disparities

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1 Abstract

Residential, racial segregation defines the life chances of Black Americans and White Americans, but in potentially different ways. However, whether these associations are the result of the residential environment or is confounded by educational and economic inequalities by race or unobserved heterogeneity is unclear. In this paper, I use data on early 20th-Century birth cohorts in the 1940 Census and exogenous variation from railroads built prior to urban development to estimate the causal effect of Black-White segregation on longevity. I show that segregation is significantly and substantively consequential for Black and White longevity. A change from perfect integration to perfect segregation decreases Black longevity by 5.9 years and increases White longevity by 3.4 years. By educational status, segregation is especially beneficial for longevity among highly educated White Americans at as much as 4 years. Conversely, the absence of an education gradient to segregation effects further supports the finding that segregation plausibly, causally affects longevity.

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2 Introduction

Black-White racial segregation is a defining feature of American cities both because of its persistence across space and time but also because of its exceptionally high levels in comparison to the segregation of other ethnoracial groups (Elbers 2021; Frey and Farley 1996). Historically, segregation peaked in the 1960s and 1970s (Logan and Stults 2022; Tolnay n.d.) and has been declining steadily since then, however average levels of Black-White segregation are still at least 10 points higher than any other pairwise ethnoracial group comparison (Elbers 2021). Trends and patterns of segregation in the United States indicate that, in both contemporary and historical perspective, racial segregation is a demographically significant physical manifestation of racial inequality (Massey and Denton 2003).

In the context of high levels of racial segregation, Black-White disparities in health, longevity, and socioeconomic resources remain persistently wide (Harper, MacLehose, and Kaufman 2014). Segregation profoundly shapes the contexts of exposure to residential environments for Black and White Americans in countervailing ways. For instance, segregation leads to concentrated exposure to poverty, violent crime, and pollution for Black Americans in ways that impact their health and potentially protects White Americans from these exposures in ways that are health advantaging (Light and Thomas 2019; Massey and Denton 2003; Wodtke, Ramaj, and Schachner 2022). Because of compelling processes outlined, researchers have linked racial segregation to health outcomes (Williams and Collins 2001).

While studies of the association between segregation and health find that segregation negatively affects Black Americans' health, it is unclear whether poor health outcomes are the result of living in a segregated place, or the individual characteristics of populations living in there. On one hand, the effects of segregation could reflect racial differences in poverty status, income, and education which affect health (Breen 2024; Hayward et al. 2000). Structural racism plays a profound role in shaping the characteristics of the neighborhoods that

Black Americans live in and economic and health outcomes ([Brown and Homan 2024](#); [Collins and Williams 1999](#); [Hendi 2024](#); [Krysan and Crowder 2017](#)) which makes disentangling the role of segregation challenging. Historical and contemporary effects of legal racial inequality such as Jim Crow ([Wrigley-Field 2024](#)), housing discrimination ([Rothstein 2018](#); [Taylor 2019](#)), and redlining are prominent ways that structural racism affects Black Americans' health outcomes ([Graetz and Esposito 2023](#)) and neighborhood residence. On the other hand, associations between segregation and health could merely reflect group differences in education, human capital, and economic resources because of their status as fundamental causes of health ([Micheltore and Rich 2023](#); [Montez and Bisesti 2024](#); [Phelan et al. 2004](#)). If segregation affects health and longevity, policies promoting racial integration may help close profound racial gaps, however if these associations are driven by educational and economic attainment, then strategies to change residential environments would be ineffective at improving population health outcomes.

In this paper, I study whether the relationship between racial segregation on longevity is causal and, after accounting for effects of segregation, whether these effects persist across the education gradient for Black and White Americans. To isolate the effect of segregation on longevity, I use data from the 1940 Census linked to NUMIDENT death records and an instrumental variable strategy that uses variation from 19th-Century railroad lines as an instrument for racial segregation ([Ananat 2011](#)). Longevity is an important aspect of population health because it is a marker of individual life chances and a cumulative summary measure health status across the life-course ([Montez et al. 2020](#); [Weber 1922](#); [Wrigley-Field 2024](#)). Comparing the effects of segregation by educational status serves as an additional way to differentiate individuals to explore whether segregation effects have an economic patterning to them – that is, if there is no educational gradient to segregation effects, then this provides further evidence toward segregation as the ultimate factor shaping longevity. Results indicate that segregation decreases Black longevity by 5.8 years and increases White

longevity by 3.4 years overall. Yet, the positive effects of segregation increase with greater education for White Americans but for Black Americans, there is no education gradient. These results emphasize the importance of the effects of structural racism on health and suggest that while segregation and residential separation from White Americans negatively affects Black health, it is by that exact mechanism that it positively affects White health.

3 Background

3.1 Prior Work on Racial Segregation and Health Outcomes

Because of the exceptional nature of Black-White racial inequalities in health ([Hayward et al. 2000](#); [Wrigley-Field 2024](#)), wealth ([Derenoncourt 2022](#)), education ([Montez and Bisesti 2024](#)), and life expectancy ([Hendi 2024](#)), and the historical persistence of segregation, scholars have drawn links between Black-White health inequalities and unequal residential arrangements. The residential separation between Black and White Americans is thought to be a physical manifestation of racial inequality and a driving force behind persistent inequality because of the stark socioeconomic and resource differences between Black and White neighborhoods ([Blokland 2008](#); [Sharkey 2013](#)). Early work by W.E.B. Du Bois documented these differences in Philadelphia as early as the 1900s ([Du Bois 1899](#)) and contemporary patterns continue to have implications for the concentration of poverty in Black neighborhoods ([Massey and Denton 2003](#)) and decreased work and education opportunities ([Wilson 1996](#)) which lead to resource disparities exacerbating health inequality ([Link and Phelan 1995](#); [Phelan and Link 2015](#); [Rothstein 2018](#)).

Prior work on segregation and outcomes has shown that segregation may negatively impact Black Americans' health. Using an instrumental variable strategy introduced by Ananat ([2011](#)) leveraging exogenous variation in pre-settlement railroads, Austin, Harper, and Strumpf ([2016](#)) found that segregation decreased the birthweight

of Black children by 2.8 grams and decreased the birthweight of White children by .63 grams. Similarly, Vu, Green, and Swan (2024) also study effects on birthweight using the same instrument and find negative impacts of segregation on the birthweight of Black children. Their results indicate that the primary mechanisms that drive these outcomes arise from disparities in access to prenatal care for Black women, structural racism, and food insecurity. In contrast, Smith (2024) found that in some cases, segregation may be beneficial for Black mental health, though these processes may be distinct from those affecting physical health and longevity. Finally, the study that this paper is most closely related to is Hendi (2024) that shows that segregation is responsible for a 16 year increase in the Black-White life expectancy gap for men and a 5 year increase for women, based on period life expectancy estimates between 1990-2019. This study makes important contributions to the empirical literature by exploring the effects of segregation on individual-level longevity that facilitate the understanding of how segregation affects health cumulatively and across the socioeconomic gradient.

At the same time, this work also has implications for White Americans. While the bulk of studies rightly focus on health outcomes for Black Americans, segregation is a system-wide phenomenon that transcends and shapes the individual factors differentiating neighborhoods where Black and White Americans reside. Vu et al. (2024) for instance found that there was no effect of segregation on the birthweight of non-Hispanic White children but Austin et al. (2016) found that segregation decreased White childrens' birthweight significantly. Similarly, while Hendi (2024) found that overall segregation widened life expectancy gaps between Black and White Americans, there were still county-level contexts where Black life expectancy was similar or exceeded that of White Americans.

While studies have documenting the effects of segregation on Black American's health and economic outcomes has been extensively documented, important methodological questions remain about whether these associations

are causal and substantively, the processes through which segregation affects Black and White Americans. Research on the persistence of segregation over time draws attention to the complex ways in which individual preferences for neighborhood racial composition and structural sorting patterns affect where people live ([Bruch and Mare 2006](#); [Krysan and Crowder 2017](#)). For instance, it could be that patterns of residential segregation and individual health and longevity are driven due to unobserved heterogeneity, preferences, or neighborhood characteristics like poverty and environmental externalities which result in the segregation-health relationship. Moreover, research shows that racial disparities in health persist at higher economic levels ([Boen 2016](#); [Esposito 2019](#)) and that more advantaged Black populations often live in similarly segregated neighborhoods to poorer populations ([Pattillo-McCoy 2000](#)). Therefore, examining how effects vary by economic status contributes clarity about whether negative health outcomes for Black Americans are primarily the result of differences economic positions or the ability to convert resources to health for Black Americans or whether disparities are driven by structural racial inequality that transcends socioeconomic status.

3.2 Theory Linking Racial Segregation to Health

County-Level Mechanisms

In recent years, scholars have demonstrated that socioeconomic differences may not fully explain Black-White health inequality because of Black Americans' residual negative health status that persists at higher SES levels ([Boen 2016](#); [DeAngelis 2022](#); [Esposito 2019](#)). Even as Black Americans ascend the economic ladder, they experience diminished health compared to White Americans ([DeAngelis, Hargrove, and Hummer 2022](#)). One important driver of these disparities is structural racism, which systematically advantages White Americans and disadvantages Black Americans through interconnected social, legal, political, and education systems ([Brown and Homan 2024:2](#)).

To date, the majority of studies linking segregation to health outcomes have focused on between neighborhood

mechanisms, but focusing on neighborhoods neglects important macro-level pathways that are consequential for health ([Brown and Homan 2024: 3](#); [Montez et al. 2020](#)). The role of administrative units, such as the county or state, are important from a public policy perspective, because they are intrinsically linked to important political, legal, and institutional processes that affect individuals living within these boundaries. Conceptually, focusing on the county-level as the unit of analysis is congruent with segregation theory and measurement, which refer to the distributions of populations at smaller geographies within a larger geography ([Duncan and Duncan 1955](#)). Studying segregation at smaller geographies may be sensitive to neighborhood change over time as census tracts change over time and their populations shift dynamically. Despite the fact that individuals may move between neighborhoods, the experience and effects of segregation are shaped by the different populations living in inter-dependent neighborhoods. While micro-level analyses capture how proximate neighborhood effects shaped by segregation affect individual outcomes, county-level analyses captures the average effect of segregation at a structural level irrespective of neighborhood differences.

At the county level, there are a variety of theoretically relevant and compelling pathways that may shape individual health outcomes. Figure 1, describes a stylized empirical diagram relating segregation to longevity, county-level and neighborhood pathways, and relevant confounding processes that are described in this section. Counties that are more segregated have are more likely to have decreased funding for public goods and public opinion in these places indicates decreased support for redistributive policies that could boost the economic conditions of working class populations ([Chyn, Haggag, and Stuart 2023](#); [McGhee 2021](#)). More generally [Chyn and Haggag \(2022\)](#), [Gilens \(2000\)](#), and [McGhee \(2021\)](#) hypothesize the common factor of these processes is greater racial resentment in more segregated counties. Through this mechanism, greater aversion to public policy that would likely improve the economic conditions of low-income Black Americans and decreased government support would likely have differential impacts on economic and material resources

that have the potential to affect low-income populations more broadly, regardless of race.

[Figure 1 Here]

County-level mechanisms also work through the criminal legal system to affect individual outcomes through institutional and governmental mechanisms that are shaped by historical racial inequality (Brown and Homan 2024; Ray 2023; Williams and Collins 2001). Compared to any other sex-age-race group, Black men and women are at the highest risk of experiencing police violence (Edwards, Lee, and Esposito 2019). Bell (2020) demonstrates that patterns of aggressive policing and punitive criminal justice policy contexts stem from high levels of segregation. In this way policing is tied to racial composition and the differing residential arrangements between White and Black Americans. Exposure to aggressive policing in segregated areas may plausibly affect both Black and White mortality through processes of arrest and incarceration. Additionally, Light and Thomas (2019) show that the segregation also creates racialized inequalities in homicide risk both through increasing Black risk, but also mechanically decreasing White Americans' risk of victimization because of residential separation. Beyond extreme outcomes like police use of force, exposure to aggressive policing leads to a higher likelihood of criminal justice contact, which has been linked to negative population health outcomes for Black Americans (Cao, Esposito, and Lee 2025; Lee 2024).

County-level effects of segregation are also shaped by historical processes and long-run patterns of racism and stratification (Graetz and Esposito 2023; Rothstein 2018). Ananat (2011) demonstrates that, in Northern metropolitan areas, segregation led to an increase in poverty for Black Americans and decreased poverty for White Americans. Additionally, segregation increased Black-White income gaps. In contemporary contexts, Graetz and Esposito (2023) show that HOLC redlining maps, which were developed for the sole purpose of maintaining segregation, created greater economic isolation and devaluation of property values in mapped areas that have proximate implications for segregation and health outcomes through homeownership (Breen

2024). At a broad level, research on the historical legacy of segregation suggests that even in an era of decline (Cutler, Glaeser, and Vigdor 1999), patterns of segregation and racial inequality have reverberating effects over time that are important for present day outcomes.

Neighborhood-Level Mechanisms

County-level processes also have downstream effects on within-county neighborhood inequalities that shape health. In places with high levels of segregation, structural forces and individual residential mobility patterns shape residential decisions that are associated with social and environmental disadvantages and unequal opportunities across neighborhoods that stratify the neighborhood conditions of Black and White individuals (Wilson 1987).

Massey and Denton (2003) argue that segregation ultimately leads to greater concentrations of poverty, which is consequential for both economic mobility and health (Aber et al. 1997). Poverty affects health through mortality directly, but also through indirect pathways that influence how health status changes over the life course through exposure to poor residential conditions, resource deprivation, and increased exposure to illness. Moreover, the levels, timing, and duration of poverty are also consequential, particularly for health in adolescence (Brooks-Gunn and Duncan 1997; Chetty, Hendren, and Katz 2016; Sharkey 2013; Sharkey and Elwert 2011). Segregation leads to higher spatial concentrations of poverty and income segregation within racial groups (Quillian 2012), that overlap with Black and White residential patterns. Racially stratified exposure to poverty has both direct effects on Black American's health outcomes by reducing educational and economic opportunities and also indirectly benefits White Americans' health by reducing their exposure (Ananat 2011; Williams, Lawrence, and Davis 2019).

Environmental characteristics of neighborhoods are also an important driver of health (Williams and Collins 2001) shaped by both exposure as well as ultimate political processes and public policies that allowed pollution

in poor and Black areas (Elliott and Frickel 2013: 20). Black neighborhoods are frequently exposed to hazardous environmental conditions and among the highest risk of being exposed to air pollution (Lopez 2002). Black and low-income populations are also at higher risk of exposure to toxic chemicals and metals such as lead (Wodtke et al. 2022) which have direct and indirect effects on health and cognitive ability. More generally, these exposures reflect a geography of hazardous environmental exposures is racially and economically stratified, with polluting industries affecting both current sites of production as well as leaving behind remnants of emissions and toxins in disproportionately minority neighborhoods (Elliott and Frickel 2013: 6).

The neighborhood context also significantly shapes the educational opportunities and school quality of Black and White children (Wodtke et al. 2023) both through racially stratified education opportunities and school funding is tied to homeownership and property taxes. Black and White students also attend segregated schools that are shaped by the demographics of their neighborhoods and correlated with educational achievement over the life course (Goldsmith 2009). A prior study by Card and Rothstein (2007) found that all else equal, segregation was responsible for a quarter of the Black-White gap in Scholastic scores.

Despite the strong associations between segregation and health, an important differentiator of Black-White health disparities is the role of educational disparities. For instance, Black Americans have on average lower education than White Americans, however, certain mechanisms that are hypothesized to stem from segregation may affect those with lower-education and economic status, regardless of race. A large body of research shows that attaining greater education is associated with better health and longevity at an individual and population level, making it complicated to separate the effects of one's environment on their health from individual characteristics (Link and Phelan 1995; Montez and Bisesti 2024; Pickett and Wilkinson 2015). Education is an important aspect of upward mobility and social status because it is a valuable

resource that accumulates over time and shapes individuals' health through knowledge, behaviors, expansion of economic opportunities, and access to health promoting resource like insurance and healthcare (Montez and Bisesti 2024). Education also increases individuals ability to convert flexible resources into health promoting and protecting behaviors (Freese and Lutfey 2011). Those who attend schools with greater resources and funding frequently concentrated within White neighborhoods, benefit from the racially stratified geography of educational opportunity.

Because of the highly coupled relationship between segregation, neighborhood opportunity, and educational attainment, it is possible that the relationship between segregation and health could be driven by underlying educational disparities. Furthermore, even if segregation primarily shapes Black and White longevity, education may moderate this relationship in important ways.

3.3 The 1940 Context

There are a few features of the data used in this study as well as the observed period of segregation that are important to contextualize when interpreting estimates in this study. The birth cohorts used in this study from 1905-19xx survived exceptional economic hardship from the Great Depression as well as the 1918 Flu Pandemic and were members of the earliest cohorts of Black migrants from the South to the North. These circumstances reflect important selectivity that must be taken into account when situating these estimates within the literature on segregation and health.

The purpose of using the 1940 Census data for analysis are to identify individuals who were part of the 1904-1918 birth cohorts who likely complete their mortality window within the 1988-2005 time frame that the Social Security Administration public data is available for (Breen 2024; Breen and Goldstein 2022). These birth cohorts are also socially and historically significant. Members of these birth cohorts are situated in a historical period marked by intense hardship followed by prosperity. These cohorts survived the first and

second World War as well as the 1918 Spanish Flu which also had distinct racial disparities (Feigenbaum, Muller, and Wrigley-Field 2019). During the 1920s and 1930s, these birth cohorts weathered the great depression, where unemployment reached highs of 24 percent (FDR Library n.d.). Amid these major changes to American Life were the prosperity that followed these material hardships, spurred in large part by expanded opportunity resulting from the New Deal introduced by President Franklin D. Roosevelt. At the same time as these changes happened, the Great Migration of Black Americans from the South to the North that occurred between 1910 and 1970 led to a restructuring of the landscape of opportunity for Black Americans in the U.S. as reshaped race relations in complex ways (Tolnay 2003).

The time period of life that I observed segregation is also important to consider when interpreting estimates. I examine exposure to segregation that mid-to-late life, obtained and calculated based on the county individuals resided at death. This decision is to maximize geographic coverage relative to the instrument that I use to establish causality and sample size for Black Americans. Relying on birth place to calculate segregation results in an analytic sample that is both too small and includes few Black individuals, because the majority of the Black population resided in the South at birth which is outside of the area that is covered by the instrument. Yet, prior evidence suggests that mid-to-late life exposure is worthy of study. Research on the life-course dynamics of migration propensity (Bernard, Bell, and Charles-Edwards 2014) suggests that individuals are less likely to migrate after age 40 years old and when they do it is usually short distances (Molloy, Smith, and Wozniak 2011; Zimran 2022). Additionally, research on aging in place shows that older adults increasingly reside in the same neighborhoods and residential contexts for 30 or more years preceding death (Bader and Warkentien 2016). For these reasons, examining exposure to segregation in the county of death represents a wider window of exposure stretching back to mid-life. It is also true that prior to the period of segregation I observe, average segregation levels of segregation in counties in my analytic sample

were higher than the observation period. Coupled with this trend, work on neighborhood attainment and residential mobility show that the modal neighborhood transition is to one of similar characteristics ([Sampson and Sharkey 2008](#)). Thus, I conclude that estimates of exposure to segregation experienced in mid-to-late life, net of birth place characteristics, proxy for life course segregation exposure that are likely attenuated from the levels experienced earlier in life.

In sum, taking selective characteristics into account, results should be interpreted as conservative estimates of segregation on longevity that are likely attenuated from what their true magnitude.

3.4 The Present Study

The goal of this study is estimate the causal effect of segregation on longevity. Longevity, defined as the number of years one lives conditional on surviving to 65 years old, is an important omnibus measure of health and life chances ([Breen 2024](#); [Weber 1922](#)). To address this question, I combine linked data from the 1940 Census with data on 19th Century railroad lines in counties in North America belonging to metropolitan areas that had populations greater than 100,000 people and populations of Black residents greater than 1,000 ([Ananat 2011](#)). I use these rail lines as an instrument for residential segregation to identify a causal effect of segregation on health. With these empirical and theoretical considerations in mind, the present study addresses two research questions.

- (1) What is the effect of county-level racial segregation on Black longevity?
- (2) What is the effect of county-level racial segregation on White longevity?
- (3) How do the effects of segregation on longevity vary by educational status for Black and White Americans?

4 Data

4.1 Data Sources

To answer my research questions, I leverage data on 1904-1918 birth cohorts of persons 22-36 years old from the 1940 Full Count United States Census. I link these data with NUMIDENT death records and the Supplementary Geographic File provided by the Berkeley CenSoc project that combines geographically linked death records from the Social Security Administration (SSA) ([Breen and Goldstein 2022](#)).

The geographic data in the NUMIDENT Geographic Supplement cover deaths occurring between 1988-2005 and reliably cover deaths over the age of 65 during this period. The geographic specificity and rich covariates, harmonized over time, make these ideal for analyses of longevity. I focus on the 1904-1918 birth cohorts for two reasons.

First, their birth years and the end year of the mortality observation window allow for likely complete observation of mortality that is not biased by missing the oldest old. In other words, those born in 1904 would be 101 years old in 2005, making it safe to assume that there is not substantial missing, right-censored data on individuals who are still alive. Second, those aged 22-36 likely have completed the majority of their formal education, which allows me to rule out the possibility of confounding related to individuals who are still in the process of completing education ([Halpern-Manners et al. 2020](#)).

My final analytic sample consists of N individuals in 147 counties at their time of death. These 147 counties were matched to 121 Metropolitan Statistical Areas included in Ananat ([2011](#)) that have information on the railroad division index instrument that I use for identification and contain populations of 1,000 or more Black residents. These counties are located outside of the American South with many concentrated in the Midwest and Northeast. After filtering and restricting observations to only non-Hispanic Black and White Americans

my final sample size results in ($N = 18,004$) Black individuals and ($N = 492,579$) White individuals.

To measure segregation, I use the dissimilarity index, a measure that describes how evenly Black and White residents are distributed in sub-geographies within a county relative to the county-level racial composition, which has been used widely in empirical research (Duncan and Duncan 1955). As mentioned previously, I calculate the dissimilarity index for the county an individual resided in at the time of their death. The dissimilarity index is interpreted as the share of Black and White individuals that would have to move for the racial composition of neighborhoods to be the same as the racial composition of the county. The dissimilarity index is calculated formally as:

$$D = \frac{1}{2} \sum_i^N \left| \frac{b_i}{B} - \frac{w_i}{W} \right| \times 100$$

Where N is the number of counties, b_i and w_i are the proportion of Black and White residents in the i th census tract and B and W are the proportion of Black and White residents in the county. The 1940 Census also includes data on county of residence at birth which I compare to county at death to determine whether individuals were movers or stayers. The dissimilarities are calculated from decennial Censuses 1980, 1990 and 2000, which correspond to the year that they passed.

Covariates and Adjustment for Double Truncation

In addition to the instrumental variable strategy, I control for a rich set of individual and contextual covariates to strengthen the validity of the findings. In fully specified models, I control for sex, education, marital status, employment status, and the proportion of Black and White older adults in the county at the time of death, and occupation fixed effects.

Birth year fixed effects are integral to reducing bias in mortality analysis with truncated data. It is well

known that the NUMIDENT data suffer from the problem of double truncation that refers to the left and right censoring of the data (Breen and Goldstein 2022). Goldstein et al. (2023) show that when not accounted for, doubly truncated observation windows result in biased estimates of longevity disparities toward 0 and mean age at death is higher. This means that, for those born in 1904, we observe them between the ages of 84-101 years old and for those born in 1918 we observe them between the ages of 70-87 years old. Following recommendations in (Goldstein et al. 2023), I include birth year fixed effects to account for double-truncation. It is also possible that there is important unobserved heterogeneity in early life experiences and residential environments common to one's place of birth that could bias estimates of segregation on longevity. To adjust for early life characteristics of individual birth places, I include birth county fixed effects from the geographic supplement in the NUMIDENT to address early life exposures.

Despite the strengths of these data there are important limitations. In terms of record matching algorithms, matches between the 1940 Census and the NUMIDENT may favor more advantaged Black Americans, resulting in estimates of the effect of segregation that are conservative. These analyses are also conditional on having no missing data in the NUMIDENT geographic supplement, which adds an additional layer of selectivity. Substantively, doubly truncated data is also a limitation because, while it is possible to recover estimates of longevity disparities, the narrow time window limits understanding whether longevity outcomes are primarily caused by earlier or later exposures to segregation (Lleras-Muney, Price, and Yue 2022). To address these representation issues, I use post-stratification weights developed by Osborne (n.d.) that re-weight CenSoc NUMIDENT counts to year-specific distributions of deaths to adjust for possible selection. Table 7 in the appendix compares the analytic sample used in this paper to the full NUMIDENT population and these populations are, on average, comparable though my analytic sample is positively educationally selected and live in slightly more segregated counties.

5 Methods

5.1 Identification and Estimation

Research has established that there are significant barriers to drawing causal inferences from observational data on the relationship between segregation and longevity. One strategy that has been used in prior research, is to find an instrumental variable that induces exogenous variation in segregation, but does not plausibly affect the outcome of interest ([Ananat 2011](#)). To identify a causal effect of segregation on longevity, I use an instrumental variable (IV) strategy first developed by ([Ananat 2011](#)) that relies on quasi-random, spatial organization of 19th-Century rail lines that were laid before urban development. This railroad division is summarized by the Railroad Division Index (RDI) that measures the degree to which a city is more or less divided up by railroad tracks, with higher values indicating more division ([Ananat 2011](#)). The RDI is calculated as follows:

$$RDI_c = 1 - \sum_{r \in c} \left(\frac{area_r}{area_c} \right)^2$$

Where $area_r$ are smaller geographic units that intersect with a railroad and $area_c$ is the total area of a larger geography (e.g., census tracts within a metro-area). The RDI is squared and the inverse of this measure which is analogous to the herfindal index ([Ananat 2011](#)).

The intuition behind the RDI as an instrument is that if a county is more divided up by railroads then those counties' populations will be more divided and will be mechanically more segregated, however it is reasonable to believe that variation in the density of railroads is likely not associated with overall longevity. The identification intuition arises through the interaction between topological features of the land where 19th Century railroads were laid and individual settlement in the 20th-Century. Historical evidence suggests that

physical barriers have long been used by White Americans to separate themselves from Black populations in coordinated ways ([Roberto and Hwang 2017](#)), thus, exogenous variation in railroad division may facilitate greater segregation both through mechanical processes as well as via ready-made barriers that White Americans could use to segregate.

When evaluating the plausibility and relevance of this instrument, it is important to do so in the context of IV assumptions. (1) the instrument is exogenous and relevant (this cannot be tested formally), (2) strength of the first stage relationship (the IV must substantially affect the endogenous variable), and (3) monotonicity (the IV only influences the level of segregation in a county in one direction).

[Figure 2 Here]

Empirically, I estimate the relationship between segregation and age at death, conditional on living to 65, in the county of residence at death. I estimate the following regression via two-stage least squares estimation, stratified by racial group ([Blandhol et al. 2022](#)), and include robust standard errors.

$$Y_i = \beta_0 + \beta_{Segregation} + \gamma_{birth-year} + \lambda_{birth-county} + \beta X_{controls} + \epsilon_i$$

Provided that these assumptions are satisfied, IV estimates of a regression of longevity on segregation are interpreted as the local average treatment effect (LATE) ([Imbens and Angrist 1994](#)), or the treatment effect amongst those living in counties where segregation increased as a result of greater railroad division.

When considering the validity of the IV there are a few important considerations beyond assumptions. First, railroad division does not imply commensurate exposure to railroads and associated externalities. That is, larger values of the RDI do not mean that there are more rail lines. The RDI refers to a summation of the proportion of census tracts within a larger geographic unit that intersect with a railroad, the RDI depends on

both the size and density of census tracts within a place as well as the number of rail lines. In some cases there may be very large and dense area with a single rail line that only intersects with a small subset of census tracts, resulting in a lower RDI, whereas a less dense place with a single rail line and larger census tracts would have a higher RDI because of the greater number of intersections, even though each context only has one line. In this way, the construction of the RDI underscores how identification results from the idiosyncrasies of geography where the tracks were laid. To quantify whether more railroad exposure could affect estimates, I run additional analyses with a second instrument for railroad track length to adjust for the amount of exposure to railroads across counties (See Table 5 in Appendix) and the results remain unchanged for migration and are qualitatively similar for Black Americans and slightly Attenuated for White Americans (See Appendix). ([Blandhol et al. 2022](#)).

A second concern might be that the railroad division might result in negative environmental exposures that may also affect health. For example, railroad division may lead to pollution resulting from coal burning that is hazardous to health or the structure of rail lines may be such that they create disparities in access to health protecting spaces or hospitals. This is likely not of significant concern for a few reasons. Firstly, the manner in which the railroads were laid and the time period that they actively polluted via burning coal were significantly before the period of exposure to segregation that I examine. This suggests that the etiological period for exposure and the potential health impacts to individuals do not overlap. Nevertheless, there could be negative environmental exposures associated with the RDI that remain in the soil or air over time, but these effects on longevity are likely small. See **Appendix: B** for a further discussion of environmental exposures.

Despite limitations, to date, the RDI has been used effectively in prior studies to examine how segregation effects poverty, birth weight, variation in life expectancy, and infant mortality ([Ananat 2011](#); [Austin et al.](#)

2016; Hendi 2024; Vu et al. 2024). In the context of trying to recover estimates as close to causal as possible, randomized experiments manipulating individual residential contexts are not feasible nor ethical.

In summary, this empirical strategy addresses critical endogeneity and selection issues in estimating the relationship between segregation and longevity, stemming from individual preferences and structural constraints and confounding the relationship. Based on the assumption that the railroad division only influences longevity through segregation, estimates can be interpreted as causal. Moreover, interpreting these results in context, the nature of the selectivity of the 1904-1918 birth cohorts, the selectivity of examining life expectancy over 65, and the selectivity in how individuals are matched from the 1940 Census to NUMIDENT point toward these estimates being conservative and attenuated. For Black Americans, in particular, descriptive statistics indicate that this sample is advantaged along death age and education, suggesting that effects of segregation are also likely attenuated from what the total population effects would be. In an ideal world with complete data coverage, estimates of the effect of segregation on longevity would likely be higher than the results presented in this paper.

6 Results

I first describe the characteristics of the sample in Table 1. As previously discussed, education differences between Black and White Americans are evident. In this sample, Black Americans had an average of 8.31 years of education compared to White Americans who had 10.73 years. In terms of residence, Black and White Americans also resided in counties with varying levels of segregation. The average Black American’s dissimilarity in the county at death was 68.9 while the average White American’s was 58. While the sample for Black Americans is much smaller sample size for Black Americans is due to the conservative nature of the matching algorithm that is used to link 1940 Census data with CenSoc NUMIDENT death records that are

shown to be more conservative for Black Americans ([Abramitzky, Boustan, and Eriksson 2012](#)) resulting more advantaged matches by economic status, education, marriage, and by extension, health.

[Table 1 Here]

Figure 3 describes Black-White racial segregation over time in the Northern counties included in my analytic sample from 1940-2000. The solid black line represents the average dissimilarity over time and grey-dashed bars represent the individual trajectories of counties over time. It is immediately evident that segregation was very high between the 1940s-1960s and then declined post-1970 at a modest rate. Counties in this sample have a slightly higher dissimilarity on average than the national dissimilarity index results reported in Elbers ([2021](#)). It is also apparent that several counties became more segregated in response to the great migration of Black Americans from the South to the North between 1940-1960 and aligns with the general insight that racial segregation became higher and more intense in this period following the inflows of Black migrants ([Derenoncourt 2022](#); [Tolnay 2003](#)).

What is notable about these changes and is important to consider when evaluating the effect size of estimates, is that between 1940 and 2000, the dissimilarity in sample counties declined from 70.2 to 51.8, which represents relatively slow decline of 18.4 points over time. Therefore, a change of even a few points for the dissimilarity index is significant. Within the sample more broadly, the least segregated county had a dissimilarity of 21.8 while the most segregated county had a dissimilarity of 86.2.

[Figure 2 Here]

In the first set of analyses, I examine the association between racial segregation and longevity using OLS regression. In figure 4, I present OLS estimates of the association between segregation and longevity by different stages of sample filtering to demonstrate how patterns change by different definitions. Table 2 describes the estimates for Black and White Americans living in the whole U.S., then those living in counties

within Northern states, then finally those living in my sample counties which are confined to the North and conditional on having 19th Century railroad lines. For all regression estimates in figures, I present estimates reflecting the difference between the minimum and maximum segregation observed within the sample (66 points) while results in regression tables describe the point estimates for a single unit increase.

Results show that segregation results in positive and substantive longevity decreases for both Black and White Americans, however the association for Black Americans is significantly greater than for White Americans. Adjusted models control for a full set of covariates and fixed effects, described in Appendix Table 3. In my analytic sample, a one unit increase in dissimilarity, the longevity for Black Americans decreased by .16 years and decreased by .12 years for White Americans. However, because OLS estimates of the association between segregation and longevity are biased, the magnitude of the coefficients is likely not accurate.

[Figure 4 Here]

The second set of analyses (Figure 5 Panel 2) examines causal estimates of the effect of segregation on longevity using two-stage least squares instrumental variable regressions. First, I begin by discussing the instrument. First, the assumptions are that (1) the RDI impacts longevity only through segregation. It is possible that rail lines could affect other population outcomes (e.g., racial composition or population size) that may affect longevity and segregation through unobserved pathways. Figure 2 indicates that the association between the RDI and population characteristics support the conclusion that the RDI does not affect longevity through population characteristics.; (2) In Figure 3, I demonstrate that the RDI is highly predictive of segregation. The first-stage F-statistic tests that test the strength of the instrument and segregation and indicate a strong and significant F-test of 36 (Table 2 in Appendix) ([Angrist and Kolesár 2021](#)). and (3) the RDI does not induce some cities to become less segregated as a result of more division.

Based on the results, estimates demonstrate a clear longevity penalty for Black Americans and in contrast

to OLS estimates, a longevity advantage for White Americans. Segregation resulted in an decrease in Black longevity by -5.8 years and an increase in White longevity by 3.3 years. What is notable about the comparison between the OLS and the IV estimates, is that, while the OLS estimates overstate the negative effects of segregation for Black Americans and reverse the sign of the coefficient for White Americans, the resulting longevity disparity between Black and White Americans is also understated at -4.3 years compared to the 9.1 year disparity compare to instrumental variable estimates. This suggests that OLS estimates bias both the sign and magnitude of Black-White longevity gaps.

Next, I turn to the second part of my analyses that examines how the effects of segregation vary by education levels. As mentioned earlier, if we believe that the effects of segregation primarily drive individual longevity outcomes for Black Americans, rather than individual-level characteristics, then education serves as a test of this hypothesis by providing information on the degree to which segregation effects follow well known education-longevity gradients.

In Figure 5, I plot the results from an interaction between segregation and years of education, stratified by racial group. I display these effects graphically following recommendations in Mize (2019). Firstly, for Black Americans, an interaction between segregation and education level is not significant based on an F-test between models. This suggests that across the education gradient, segregation leads to significant reductions in longevity for all Black Americans. Conversely, we see the opposite relationship for White Americans. Across educational status, education significantly increases the positive effects of segregation on longevity with effects ranging from 1 to 4.4 years depending on education level, with the only exception to this relationship being those who have 0 years of education. Importantly, while the effects of segregation on White longevity are positive for all levels of education, lower-educated White Americans do not receive the same longevity benefits as higher-educated groups, suggesting that there is educational stratification in the degree to which

White Americans benefit from the effects of segregation.

[Figure 5 Here]

Overall, the broad takeaway from these analyses by education is that segregation is at least positive and beneficial for the longevity of White Americans across all education levels and harmful for Black Americans regardless of education.

Across models, these estimates remain robust to additional controls and estimates incorporating a second instrumental variable to control for the length of railroad tracks that individuals live around (Appendix Table 5). Additionally, main estimates use individual-level statistical weights provided by CenSoc to improve the generalizability of the sample compared to the 1940 U.S. population. Following the same procedure as Breen (2024), I also ran analyses on the unweighted sample as a robustness test and results remain robust to weighting methods.

7 Conclusion

In this paper I demonstrate that residential segregation is responsible for substantively and statistically significant longevity penalties for Black Americans. By education status, results indicate that there is not an education gradient to segregation effects for Black Americans, which further supports a causal relationship between segregation and health. For White Americans, the reverse is true. Segregation results in positive and significant effects on White longevity, which likely stems from reduced exposure to crime and poverty that is overwhelmingly concentrated in Black neighborhoods within segregated spaces (Light and Thomas 2019; Massey and Denton 2003). These results emphasize the importance of the effects of structural racism on health and suggest that residential separation of White and Black Americans yields disparate and unequal effects on Black health, but positive effects White health. Yet, a the presence of a positive and significant

education gradient for White Americans, with effects on longevity especially pronounced for the highly educated, points toward additional nuance needed in the interpretation of these estimates.

Progress toward redressing racial inequality more generally is often framed as a zero-sum relationship between Black and White Americans, whereby White Americans must pay an expense for equality. These analyses suggest that evidentiary and empirical basis for these claims are questionable. While it is the case that White Americans that the majority of White Americans benefit from segregation, within-group disparities in longevity by education, demonstrate that there is significant heterogeneity in these effects. Comparing the lowest to the highest educated, the disparity is 3 years (or 38 percent of the total Black-White longevity disparity). To put this number in context, the total White longevity advantage, regardless of education, is 3 years, meaning that the longevity advantages among White Americans vary across education status at a similar magnitude to the average effect. Furthermore, though this is a small group (.4 percent), White Americans with less than 2 years of education are estimated to have no positive return to segregation.

While this study makes progress toward disentangling causality from observed associations between segregation and longevity, there are limitations to this methodology worth discussing. Firstly, while instrumental variable (IV) methods provide purchase on answering causal questions with cross-sectional data, they require strong assumptions, many of which are fundamentally untestable and rely on careful reasoning and domain knowledge of the case. For instance, while results support the relevance assumption of IV, I can only reason about the degree to which the exclusion restriction holds. For instance, evidence suggests (Table 6) that the instrument does not affect other population characteristics (e.g., the population size and share of Black and White residents) which may affect other population patterns which plausibly affect health. Yet, it is also possible that the instrument may affect unobserved pathways that relate to longevity in some way.

In contrast to other methods for studying the effects of segregation on health, such as sibling fixed effects,

IV methods offer an advantage because they do impose restrictions on where variation comes from (e.g., within family) which may result in estimates that have high internal validity but represent a highly selective population ([Engzell and Hällsten 2024](#)).

- Racial segregation embeds racial inequality in places which prevent diverse interactions and inhibit potential socioeconomic status (SES)-based solidarity ([Bell 2020](#); [DuBois 1935](#)). Connect to work by Daniels ([1975](#)) on White Americans paying a premium to segregate.
- Conclusions from these findings are consistent with prior work examining how segregation affects both Black and White Americans. Light and Thomas ([2019](#)) conclude that one explanation for the persistence of segregation is the reduced homicide risk that White Americans receive by living in segregated neighborhoods. These results confirm that similar longevity advantages among White Americans as a result of segregation are produced and longevity disparities are likely shaped by racial disparities in crime exposure amongst other things.
- Results for Black and White Americans represent a particular birth cohort that lived during a period of significantly greater racial inequality. Since then, race relations have changed significantly and income inequality has risen sharply. It is also the case that access to education has expanded significantly and effects of segregation by education in more recent years may be different merit further study.

Table 1: Descriptive Statistics of Analytic Sample (1904-1918 Birth Cohorts)

	Black		White	
	Mean	SD	Mean	SD
Age at Death	82.63	5.87	82.98	5.56
Male	0.44	0.50	0.42	0.49
Education (Years)	8.31	3.47	10.73	2.99
County D at Time of Death	0.69	0.13	0.58	0.15
Married	0.61	0.49	0.54	0.50
Decade of Death	1992.28	6.22	1993.03	6.09
Birth Year	1913.63	3.72	1914.00	3.52
Pr. Black 65+	0.08	0.02	0.07	0.02
Pr. White 65+	0.15	0.03	0.14	0.03
N	18004		492579	

Notes: This table describes summary statistics for the analytic sample of individuals aged 20-36 at the time of the 1940 Census. The analytic sample is filtered based on two conditions: 1) no missing birth and death county information in the geographic supplement file; 2) no missing county-level dissimilarity information or and coverage with the RDI instrument for segregation at time of death.

Note that age at death are unadjusted and thus not readily interpretable because of the doubly truncated nature of the NUMIDENT death window.

8 Appendix: A Main tables and figures

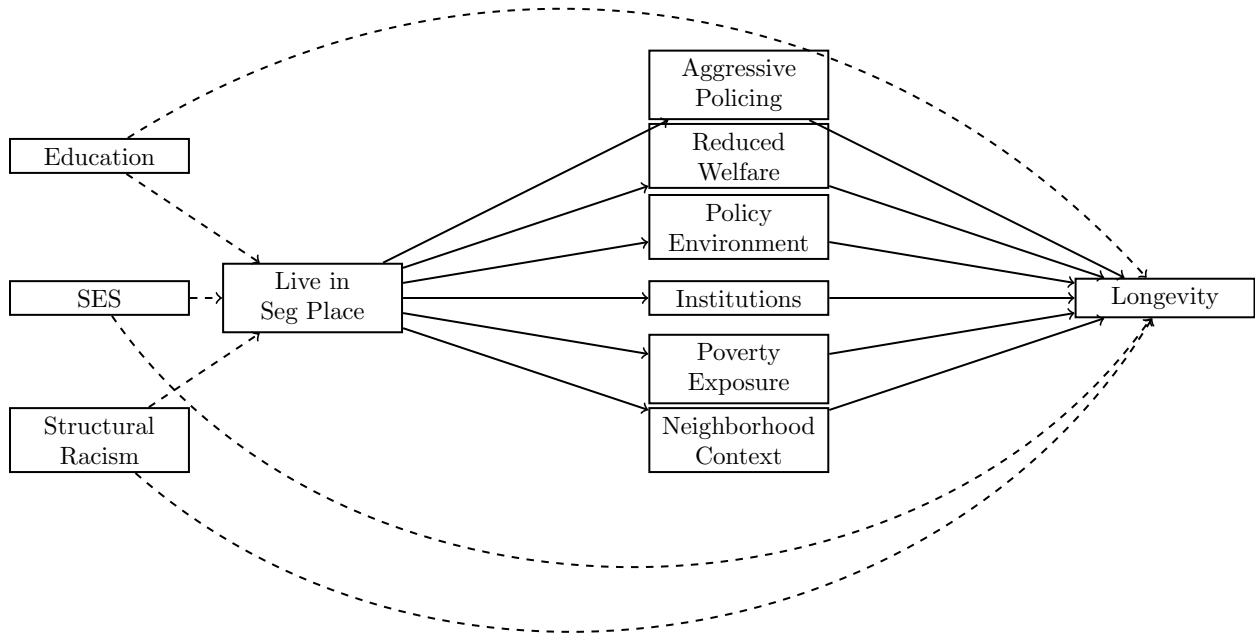


Figure 1: Empirical Diagram Relating Segregation to Longevity

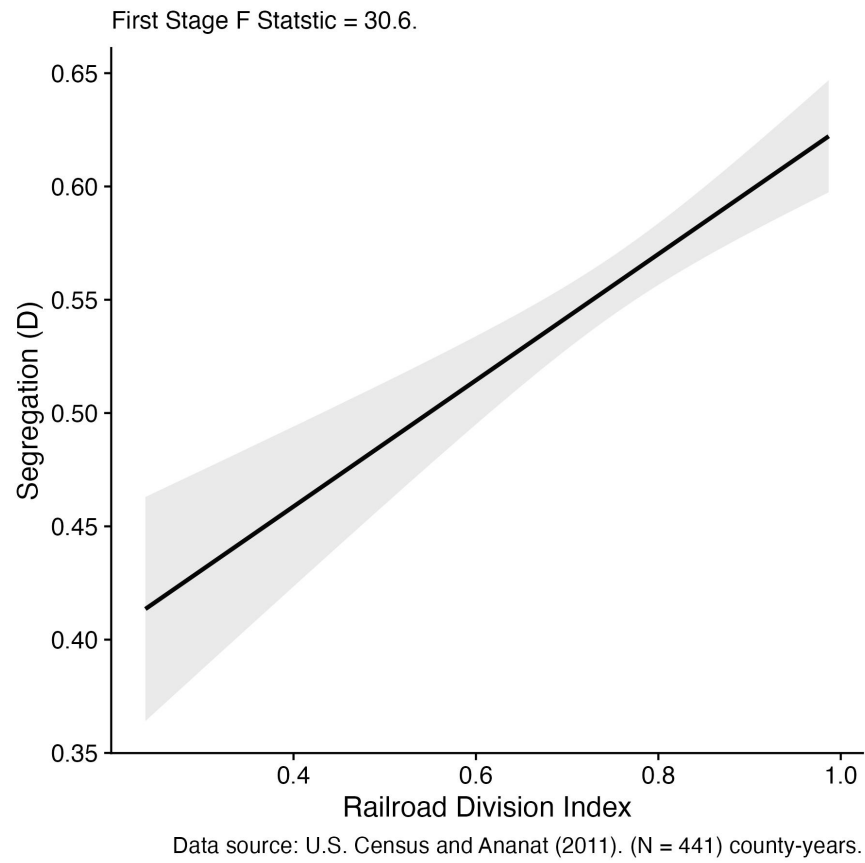


Figure 2: Association Between RDI and Segregation

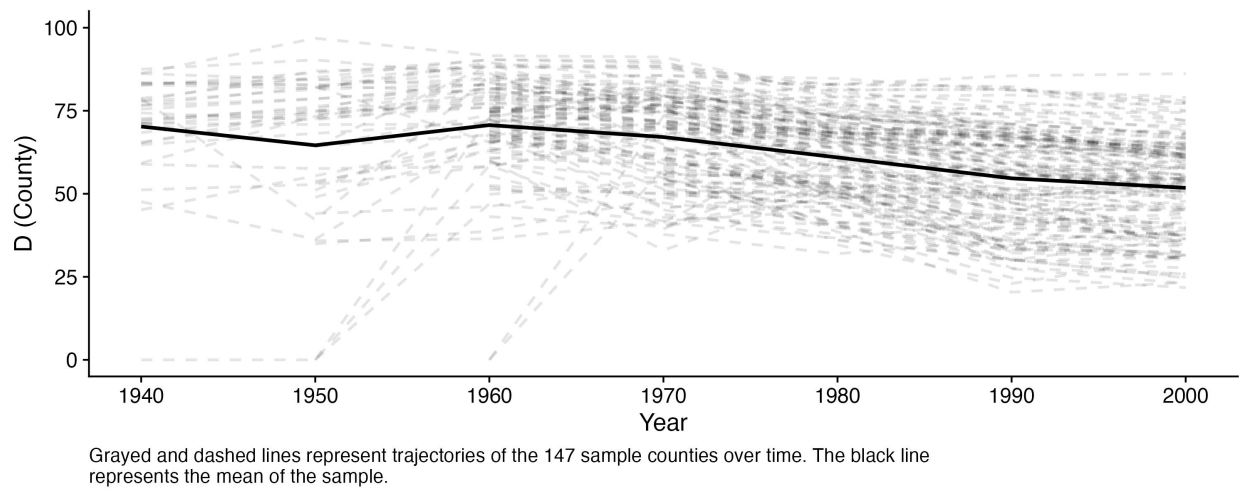


Figure 3: County-Level Dissimilarity in Sample Counties Over Time

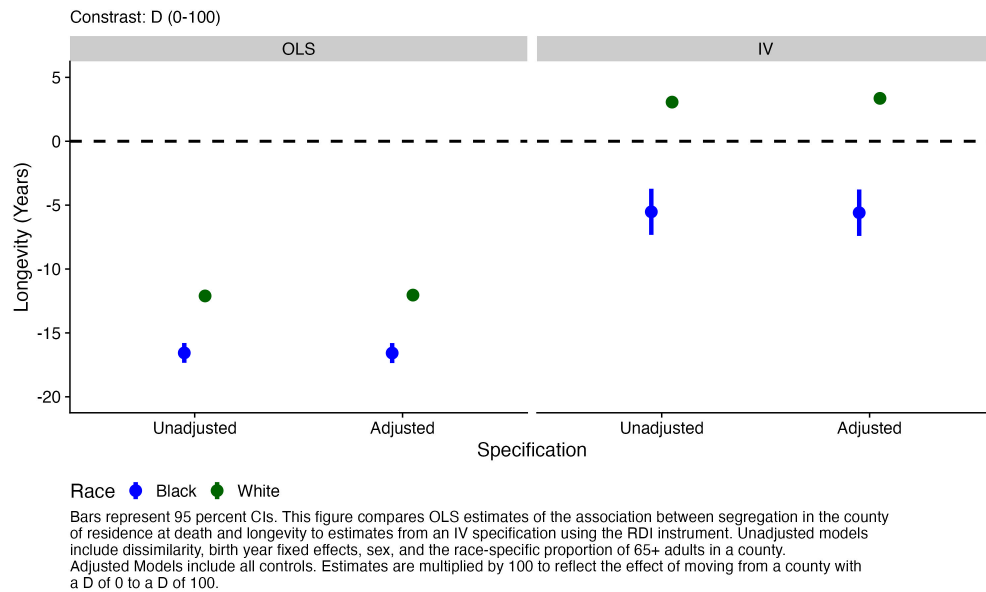


Figure 4: OLS and IV Estimates of the Effect of Segregation on Longevity

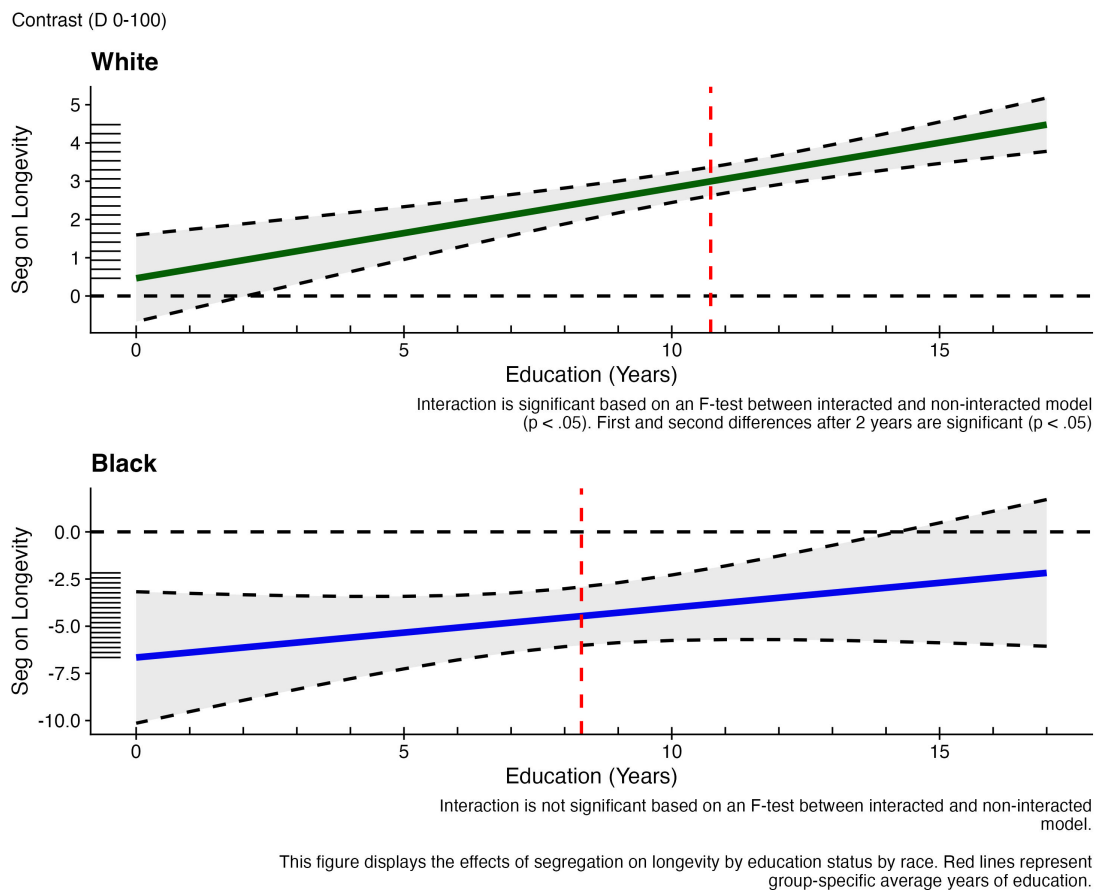


Figure 5: The Effect of Segregation on Longevity by Educational Status

Table 2: First Stage Relationship

	(1)
Constant	0.35*** (0.04)
RDI	0.28*** (0.05)
Num.Obs.	441
F	36.629

+ $p < 0.1$, * $p < 0.05$,
** $p < 0.01$, *** $p < 0.001$

First stage regression of county-level D on RDI. All models use robust standard errors.

Table 3: OLS Estimates of the Association Between Racial Segregation and Longevity

	Black	White	Black (Controls)	White (Controls)
D (White-Black)	-16.56*** (0.40)	-12.11*** (0.06)	-16.58*** (0.40)	-12.04*** (0.06)
Male	-1.01*** (0.08)	-0.92*** (0.02)	-0.93*** (0.12)	-0.96*** (0.02)
Education (Years)			0.03* (0.01)	0.05*** (0.00)
Married			0.14 (0.09)	0.04* (0.02)
Employed			0.29* (0.13)	0.19*** (0.03)
Proportion White 65+		81.67*** (0.31)		81.59*** (0.31)
Proportion Black 65+	103.37*** (2.37)		103.66*** (2.38)	
Num.Obs.	18004	492579	18004	492579
FE: byear	X	X	X	X
FE: bfp	X	X	X	X
FE: OCC			X	X

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

OLS regression models are stratified by race. All models use robust standard errors and observations are weighted using CenSoc NUMIDENT post-stratification weights. Dissimilarities are multiplied by 100 for interpretation.

Table 4: IV Estimates of the Effect of Racial Segregation on Longevity

	Black	White	Black (Controls)	White (Controls)
D (White-Black)	-5.52*** (0.92)	3.06*** (0.22)	-5.59*** (0.93)	3.36*** (0.22)
Male	-1.09*** (0.08)	-0.97*** (0.02)	-0.99*** (0.13)	-1.02*** (0.02)
Education (Years)			0.04** (0.01)	0.08*** (0.00)
Married			0.15 (0.09)	0.11*** (0.02)
Employed			0.24+ (0.14)	0.24*** (0.03)
Proportion White 65+		63.38*** (0.43)		63.10*** (0.43)
Proportion Black 65+	76.53*** (3.13)		77.06*** (3.15)	
Num.Obs.	18004	492579	18004	492579
FE: byear	X	X	X	X
FE: bfp	X	X	X	X
FE: OCC			X	X
First-Stage IV F Statistic	1895.60	27290.70	-	-

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

This table presents results from Two-Stage Least Squares regression models, stratified by race, of the effect of White-Black Dissimilarity on Longevity. All models use robust standard errors and observations are weighted using CenSoc NUMIDENT post-stratification weights. Dissimilarities are multiplied by 100 for interpretation.

Table 5: IV Estimates of The Effect of Racial Segregation and Longevity (Robustness Tests)

	Migration (Black)	Migration (White)	Track Length (Black)	Track Length (White)
D (White-Black)	-5.496*** (0.923)	3.776*** (0.223)	-5.857*** (0.909)	1.359*** (0.191)
Male	-1.092*** (0.082)	-0.968*** (0.017)	-1.089*** (0.082)	-0.968*** (0.016)
Migrated	-0.466+ (0.238)	-1.442*** (0.027)		
Proportion White 65+		65.876*** (0.427)		65.435*** (0.411)
Proportion Black 65+	76.639*** (3.132)		77.347*** (3.104)	
Num.Obs.	18004	492579	18004	492579
AIC	108021.8	2908562.5	107946.8	2888325.2
FE: byear	X	X	X	X
FE: bfip	X	X	X	X

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

This table presents IV regression models, stratified by race, of the effect of Black-White segregation and longevity. All models use robust standard errors. Dissimilarities are multiplied by 100 for interpretability. The first two models adjust for migration (whether an individual moved from their birth county to a different death county) and the second set of models includes an additional instrument for railroad track length.

Table 6: Association Between RDI and Population Characteristics (1980-2000) Pooled

	Pr. Black	Pop.	N Black	N White
RDI	0.054 (0.028)	-65538.680 (139168.077)	76136.013 (40429.844)	-11157.867 (91764.920)
Num.Obs.	441	441	441	441

This table displays falsification tests of the RDI. If the RDI is to be a valid instrument, it should not predict other population characteristics. If the RDI did predict other population characteristics, then the instrument may affect outcomes through other pathways than segregation.

Table 7: *

Summary Statistics	Comparing Full NUMIDENT Sample to		Analytic Sample	
	NUMIDENT Geographic Supplement		Analytic Sample	
	N	Mean or Pr.	N	Mean or Pr.
Age at Death	1942196	83.14	510630	82.97
Birth Year	1942196	1913.89	510630	1913.99
Decade of Death	1942196	1993.13	510630	1993.00
Black	1942196	0.05	510630	0.04
Male	1942196	0.41	510630	0.42
Married	1942196	0.57	510630	0.54
Education (Years)	1942196	10.25	510630	10.64
County D at Time of Death	1942196	0.53	510630	0.59
Pr. Black 65+	1942196	0.08	510630	0.07
Pr. White 65+	1942196	0.15	510630	0.14

This table compares the Analytic Sample conditional on instrument coverage with the full CenSoc NUMIDENT Geographic Supplement sample.

9 Appendix: B

As noted earlier, there are notable limitations that could threaten identification and the strength of the instrument. A leading limitation is the possibility that railroad division led to greater exposure to pollution through the use of coal to operate railroads leading to both contemporaneous exposure and exposure through soil. Figure 6 provides a visual timeline of when railroads were built, when they operated using coal, and the timeline that birth cohorts in this study were alive.

According to historical record, the majority of steam engines ceased their operations by the 1960s ([Stagner 1991](#)) which precedes the start of the period that segregation is observed in this study by 20 years, suggesting that contemporaneous exposures that may affect longevity are not sufficiently worrisome for these estimates. Still, there are some sample members that lived in the same county their entire lives that may have been affected.

Table 10 describes the shares of Black and White sample members who live in the same county at birth as

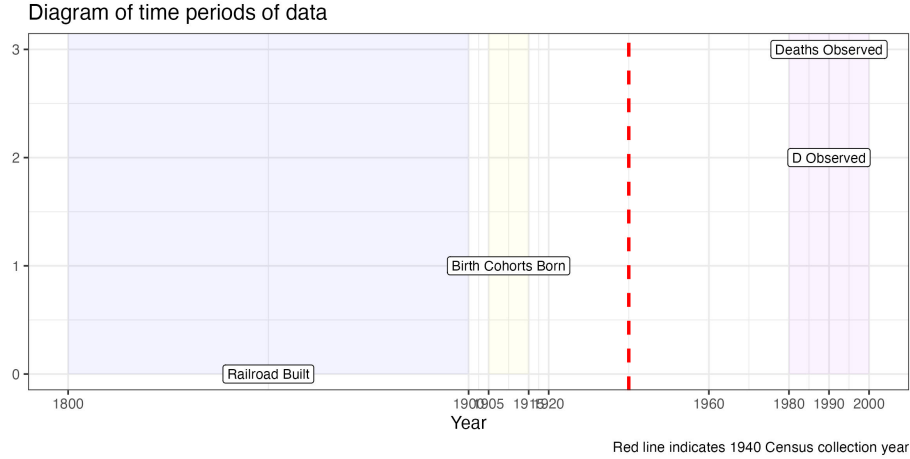


Figure 6: Data Timeline

Table 8: Count and Proportion of Sample Living in the Same County Since Birth by Race

County	Black		White	
	N	Percent	N	Percent
Does Not Live in Birth County	21681	90.25	438602	69.84
Lives in Birth County	2343	9.75	189377	30.16

death. The vast majority of Black sample members do not live in the same county that they were born in due to the great migration of Black Americans from the South to the North, while about 70 percent of White Americans migrated. These patterns indicate that if bias as a result of contemporaneous exposure to railroad coal pollution does affect estimates, it does not affect the majority of Black sample members but may affect estimates for White sample members.

It is still possible that soil pollution may affect longevity leading to a spurious association between segregation and health, however it is unclear whether the data needed to link railroads to soil pollution to test these associations exist. Given these circumstance, this is a limitation of the current study.

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